

Private 5G with adaptive network slicing enables low-latency remote control of industrial robots, as demonstrated in recent trials.

Manufacturing × Remote and autonomous operations × Private 5G / LTE · OS077 v6 · refreshed 2026-08-25

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
74 is the world moving	88 can we play, can we win	HIGH 0 named in the evidence	€3.2bn partial evidence · per year	NEXT pilots published no volume pr...	L0 46 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

Private 5G with adaptive network slicing enables low-latency remote control of industrial robots, as demonstrated in recent trials. This opportunity targets manufacturers with mobile robots or AGVs that need deterministic, low-latency wireless connectivity for remote and autonomous operations. It exists now because private 5G has moved from pilots to early enterprise deployments, and the technology for slicing and latency optimization has matured.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€4.5bn €419m – €54.4bn	€3.2bn €300m – €39.0bn	€193m €18.0m – €2.3bn	partial
Observed: tendered contract value, annualised	€88.6m €88.6m – €180m	€88.6m €88.6m – €180m	€5.3m €5.3m – €10.8m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Manufacturing (EU27_2020)	238,122 enterprises	observed	Eurostat · sbs_sc_oww	2024
Enterprises adopting Private 5G / LTE	30.0 % of enterprises (10+ employees)	proxy	Eurostat · isoc_eb_iotn2 · E_IOT1	2021
Annual value of one engagement	€380k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Size-weighted buyer base	39,220 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	6.0 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Internet of Things by NACE Rev. 2 activity series E_IOT1 is used as a proxy for Private 5G / LTE: No private-network series; industrial IoT use as proxy. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

Observed: tendered contract value, annualised

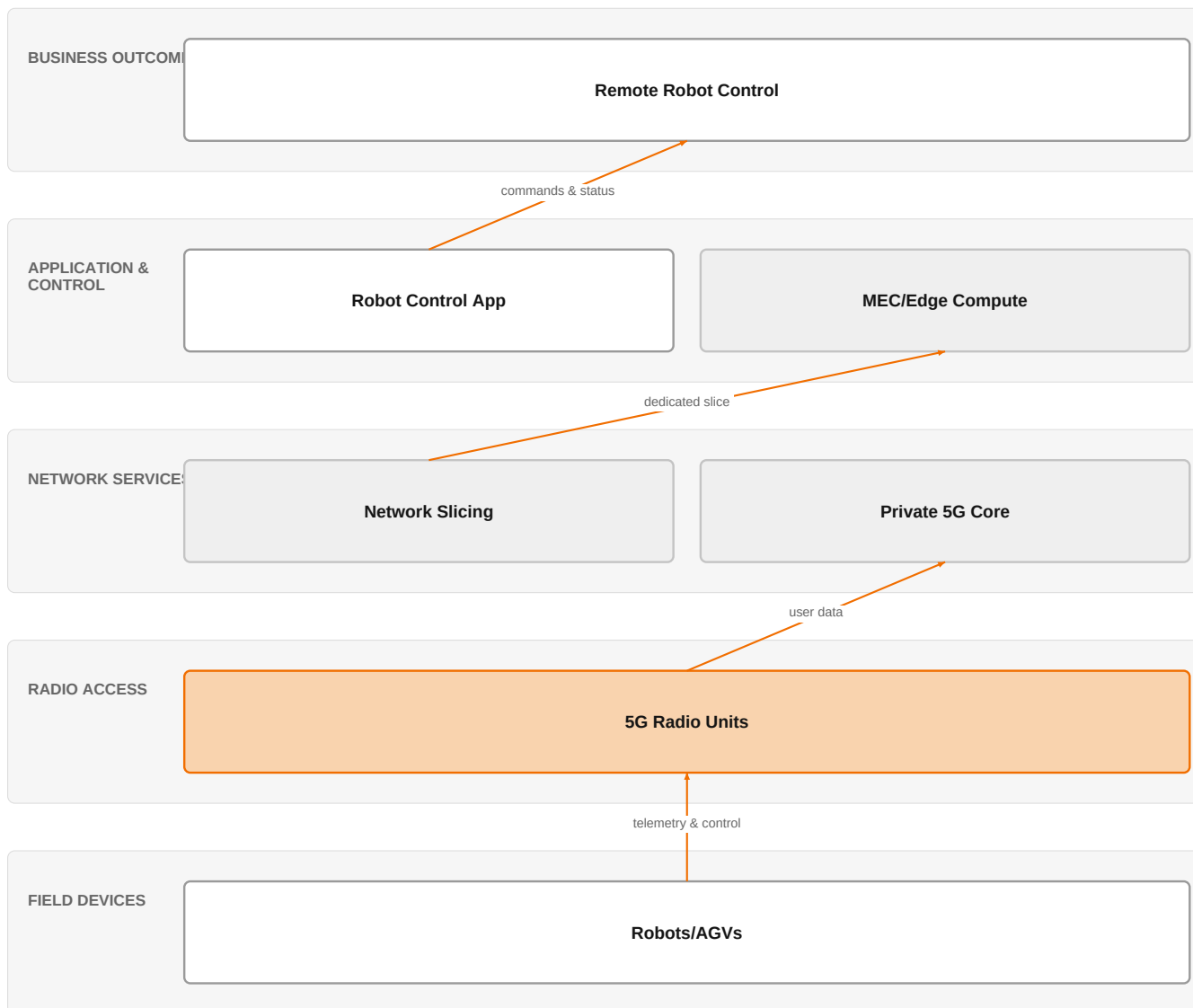
Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€88.6m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Assumed obtainable share of the serviceable market	6.0 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 90 days of TED coverage and scaled to a year (x4.1); a short collection window makes that scaling rough. 30 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a

market size.

The solution, in one picture

Private 5G for Robot Control



This diagram shows how a private 5G network with slicing provides deterministic low-latency connectivity between the control application and robots, enabling reliable remote operation.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Recent IEEE demonstrations show teleoperated driving via private 5G networks with remotely-driven robots, proving the technical feasibility of low-latency remote control. Cross-layer latency optimization for URLLC services in 5G MEC networks using adaptive network slicing addresses the challenge of maintaining strict deadlines in dynamic settings, which is critical for industrial automation. Network slicing within private 5G networks enables enterprises to offer differentiated QoS levels for various applications, moving from best-effort to deterministic service-oriented architectures. The large-scale deployment of dynamic terminals like mobile robots and AGVs in industrial private networks requires optimized base station deployment and coverage, as conventional methods degrade in accuracy. Integrating Time-Sensitive Networking with 5G private networks is emerging to meet the stringent delay and jitter requirements of IIoT services, supporting Autonomous Mobile Robots and dynamic workflows.

Sources: SIG-B11748E56A0F 2026 IEEE 12th Interna 2026-06-29; SIG-18012D9BCACF Springer Science and B 2026-06-19; SIG-DDBBDA6A8C40 International Scientif 2026-06-19; SIG-67F7CBD007F7 Applied Sciences 2026-05-22; SIG-FFC21218DC93 openalex.org 2026-03-12

Who buys, and why now

The buyer is typically a manufacturing operations or IT director responsible for production efficiency and safety. The pain is operational: mobile robots and AGVs need reliable, low-latency wireless control to avoid stoppages, collisions, or safety incidents. The trigger is often a failed Wi-Fi deployment, a new automation project requiring deterministic connectivity, or a safety audit highlighting wireless reliability gaps. They sign because they need a network that can guarantee latency and reliability for real-time control, which Wi-Fi cannot provide.

Why it is hot now — every claim bound to a dated source

- Teleoperated driving via private 5G demonstrates remote operation of industrial robots. [SIG-B11748E56A0F]
- Demonstration of teleoperated driving via a private 5G network with remotely-driven robots. [SIG-B11748E56A0F]
- Cross-layer latency optimization for URLLC services in 5G MEC networks using adaptive network slicing supports latency-critical applications like industrial automation. [SIG-18012D9BCACF]
- Teleoperated driving via private 5G networks has been demonstrated, enabling remote operation of robots. [SIG-B11748E56A0F]
- Private 5G networks demonstrate teleoperated driving with remotely-driven robots, indicating viable remote operations. [SIG-B11748E56A0F]
- Integration of private 5G with UWB localization supports collaborative robots in industrial settings, providing high throughput and low latency. [SIG-5E2C87F5E70E]
- Cargill is deploying private 5G across its sites to modernize manufacturing operations. [SIG-1FF16F75FD01, SIG-8B7741B3D9CF]
- Teleoperated driving via a private 5G network has been demonstrated with remotely-driven robots. [SIG-B11748E56A0F]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would deliver a private mobile network (L0) tailored to the customer's factory, leveraging its position as a Leader in the Gartner Magic Quadrant for 4G and 5G Private Mobile Network Services. The engagement would start with a site survey and network design, then deploy the private 5G/LTE infrastructure, possibly with partners like Samsung or Huawei for RAN equipment. Orange would then implement network slicing to create a dedicated low-latency slice for robot control, and integrate with the customer's existing MEC or edge computing resources. The team would include IoT and Digital experts to ensure the solution meets operational needs, and could draw on Orange Group researchers for optimization. The solution would be delivered with ISO 27001 and TISAX certifications to address security and automotive-specific requirements.

Why Orange can win

Orange can win because it is recognized as a Leader in the Gartner Magic Quadrant for 4G and 5G Private Mobile Network Services, providing a credible, proven offering. It has a pool of IoT, Digital, and Defense and security experts to support complex industrial deployments. The certifications (ISO 27001, TISAX) add trust, especially for security-conscious manufacturers. However, the assets are thin on specific industrial robot references; the named references (Colgate-Palmolive, Mondelez International, Saint-Gobain Glass) are consumer goods and glass, not heavy robotics, so Orange must lean on its general private network expertise and partner ecosystem.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Private mobile networks	offer	L0	offer addresses the use case AND provides the technology	unconfirmed
Samsung	partner	L2	partner provides the required technology	unconfirmed
Huawei	partner	L2	partner provides the required technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
TISAX	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
Gartner Magic Quadrant 2026 — 4G and 5G Private Mobile Network Services (Leader)	analyst position	SUP	analyst recognition covers this technology or use case	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Colgate-Palmolive	reference	SUP	published reference in this vertical, different use case	unconfirmed
Mondelez International	reference	SUP	published reference in this vertical, different use case	unconfirmed
Saint-Gobain Glass	reference	SUP	published reference in this vertical, different use case	unconfirmed
Skytale	reference	SUP	published reference in this vertical, different use case	unconfirmed

6 further published reference(s) in this vertical are linked to this topic and are listed in the radar.

Competitive landscape

HIGH competitive intensity (score 8.0; 0 of 8 listed players appear in this topic's own evidence)

Crowded, with heavyweight incumbents already visible in the evidence. Win on a specific differentiator or do not bid.

Competitors include telecom operator peers like BT Business, Deutsche Telekom / T-Systems, Proximus, Swisscom, Telefonica Tech, Telia Company, and Verizon Business, all selling private 5G/LTE connectivity solutions. NTT DATA, a global systems integrator, also competes and has a real-world deployment with Cargill, demonstrating its ability to modernize manufacturing operations. These competitors are strong in connectivity and have their own private network offerings; NTT DATA differentiates with integration services. The customer will compare on price, coverage, reliability, and the ability to integrate with existing IT/OT systems.

Sources: SIG-8B7741B3D9CF NTT Data 2026-02-18

Competitor	Category	Position here	Basis	Evidence
BT Business	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	sells Private 5G / LTE; active in Manufacturing; competes in Connectivity Solutions	structural	—
NTT DATA	Global systems integrator	sells Private 5G / LTE; competes in Connectivity Solutions	structural	—
Proximus	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions	structural	—
Swisscom	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions	structural	—
Telefonica Tech	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions	structural	—
Telia Company	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions	structural	—
Verizon Business	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Competitor analysis

6 of 8 competitors on this space have been read from their own published pages; 2 could not be read and are marked below.

The shape of the field

The field for private 5G in manufacturing is dominated by telecom operators and systems integrators, but most competitors' published pages do not explicitly address autonomous mobile robots or deterministic communications. The gap lies in operators that can combine network ownership with industrial expertise and proven manufacturing references. Orange's Private mobile networks offer, backed by multiple manufacturing references and a Gartner Magic Quadrant 2026 Leader position, positions it well to lead in this niche, while competitors like BT, Proximus, and Verizon focus on broader connectivity or lack specific industrial private 5G offerings.

BT Business Telecom operator peer · structural presence

BT Business presents itself as a provider of connectivity, cloud, and security solutions for UK businesses, ranging from broadband and mobile to IoT and drone services. It targets a broad set of industries including retail, banking, manufacturing, healthcare, and the public sector, with a focus on digital transformation and industry-specific challenges.

Doing here: BT Business's published pages focus on broadband, IoT, drones, and network security, with no specific mention of private 5G for autonomous mobile robots in manufacturing. Their IoT solutions target businesses with over 100 employees, and their drone offerings include connectivity and survey services, but nothing on deterministic communications for industrial automation.

How Orange differentiates: Orange differentiates through its Private mobile networks offer, which is positioned for industrial use cases like autonomous mobile robots, backed by references such as Colgate-Palmolive and Mondelez International. BT Business's pages do not show a comparable private 5G offering for manufacturing, focusing instead on broadband and drones. Orange can lead with its proven deployments in manufacturing and its Gartner Magic Quadrant 2026 Leader position for private mobile network services, while avoiding a direct comparison on drone services where BT has a specific offering.

Anchored on: Private mobile networks (L0) · Colgate-Palmolive (SUP) · Mondelez International (SUP) · Gartner Magic Quadrant 2026 — 4G and 5G Private Mobile Network Services (Leader) (SUP)

What they do better: BT Business has a strong drone service portfolio in the UK, which Orange does not directly match.

Deutsche Telekom / T-Systems Telecom operator peer · structural presence

T-Systems, a subsidiary of Deutsche Telekom, presents itself as a provider of digital services for businesses and public institutions, with core competencies in cloud services, artificial intelligence, and security. It offers managed services for IT infrastructure, including edge computing, IoT, and cybersecurity solutions, and positions itself as a partner for digital transformation across industries.

Doing here: Deutsche Telekom / T-Systems's pages emphasize sovereign AI, cloud, and cybersecurity, including OT security and drone detection, but do not mention private 5G for autonomous mobile robots in manufacturing. Their OT security solutions protect industrial OT-IT infrastructure, and they offer a drone detection system, but nothing on deterministic communications for industrial automation.

How Orange differentiates: Orange differentiates with its Private mobile networks offer, which is specifically designed for industrial use cases like autonomous mobile robots, and has published references in manufacturing (e.g., Saint-Gobain Glass, Skytale). T-Systems's pages focus on OT security and sovereign AI, but do not show a private 5G offering for manufacturing. Orange can lead with its end-to-end private 5G solution and manufacturing references, while acknowledging T-Systems' strength in OT security and sovereign cloud.

Anchored on: Private mobile networks (L0) · Saint-Gobain Glass (SUP) · Skytale (SUP)

What they do better: T-Systems has a strong OT security portfolio and sovereign AI platform, which are complementary but not directly competitive in this space.

NTT DATA Global systems integrator · structural presence

NTT DATA presents itself as a global systems integrator and IT services provider, offering end-to-end consulting, technology, and managed services across infrastructure, AI, data, and sustainability. It targets large enterprises and public sector organizations, emphasizing AI-driven modernization and global delivery.

Doing here: NTT DATA's pages describe full-stack infrastructure, AI-native solutions, and secure networking, but do not mention private 5G for autonomous mobile robots in manufacturing. Their secure networking solutions cover campus, WAN, and cloud, and they offer hybrid data center solutions for AI workloads, but nothing on deterministic communications for industrial automation.

How Orange differentiates: Orange differentiates as a telecom operator with an owned network and a dedicated Private mobile networks offer, whereas NTT DATA is a systems integrator without its own radio access network. Orange's

manufacturing references (e.g., Everise, Carso) demonstrate real-world deployments, and its Gartner Magic Quadrant 2026 Leader position underscores its credibility. Orange can lead with its network ownership and private 5G expertise, while avoiding a direct comparison on global systems integration where NTT DATA has breadth.

Anchored on: Private mobile networks (L0) · Everise (SUP) · Carso (SUP) · Gartner Magic Quadrant 2026 — 4G and 5G Private Mobile Network Services (Leader) (SUP)

What they do better: NTT DATA has a global systems integration capability and AI-native solutions that Orange does not match in breadth.

Proximus Telecom operator peer · structural presence

Proximus presents itself as a Belgian telecom operator offering fixed and mobile connectivity, devices, and ICT solutions for consumers, independents, small businesses, and larger companies. Its offerings include fiber internet, mobile subscriptions, Wi-Fi technology, cloud, cybersecurity, and collaboration tools.

Doing here: Proximus's pages focus on consumer and small business connectivity, including fiber internet, Wi-Fi, and mobile subscriptions, with no mention of private 5G for autonomous mobile robots in manufacturing. Their Business Flex+ pack includes cybersecurity and cloud solutions, but nothing on industrial private networks.

How Orange differentiates: Orange differentiates with its Private mobile networks offer, which is tailored for industrial use cases like autonomous mobile robots, and has manufacturing references such as Tricots Saint James and Aberg Connect. Proximus's pages do not show a private 5G offering for manufacturing, focusing instead on consumer and small business connectivity. Orange can lead with its industrial private 5G expertise and references, while avoiding a direct comparison on consumer broadband where Proximus has a strong local presence.

Anchored on: Private mobile networks (L0) · Tricots Saint James (SUP) · Aberg Connect (SUP)

What they do better: Proximus has a strong consumer and small business connectivity portfolio in Belgium, which Orange does not directly match in that market.

Swisscom Telecom operator peer · structural presence

Swisscom presents itself as a comprehensive ICT partner for Swiss businesses, offering integrated solutions across cloud, security, IoT, and digitalization, with a focus on enabling, driving, and protecting business operations. It emphasizes personal support, quality, and expertise, and highlights its role in helping companies navigate digital transformation, including the adoption of AI and data-driven strategies.

Doing here: Swisscom's pages describe digital manufacturing services, IoT Data as a Service, and a Swiss AI Platform, but do not specifically mention private 5G for autonomous mobile robots. Their digital manufacturing services support digitalisation of production, and they help industrial companies overcome data silos, but nothing on deterministic communications for industrial automation.

How Orange differentiates: Orange differentiates with its Private mobile networks offer, which is specifically designed for industrial use cases like autonomous mobile robots, and has manufacturing references such as Amerigo and MPSA. Swisscom's pages focus on digital manufacturing services and IoT, but do not show a private 5G offering for deterministic communications. Orange can lead with its private 5G network expertise and manufacturing references, while acknowledging Swisscom's strength in sovereign AI and IoT data services.

Anchored on: Private mobile networks (L0) · Amerigo (SUP) · MPSA (SUP)

What they do better: Swisscom has a strong sovereign AI platform and IoT data services, which are complementary but not directly competitive in this space.

Telefonica Tech Telecom operator peer · structural presence

Their published position is unread — 403 to automated clients. Nothing below is inferred on their behalf.

Doing here: Telefonica Tech's published position is unread because their site blocks automated clients (403). No activity can be confirmed from their pages.

How Orange differentiates: Since Telefonica Tech's published position is unread, Orange cannot differentiate against specific claims. However, Orange can rely on its Private mobile networks offer and manufacturing references (e.g., Colgate-Palmolive, Mondelez International) to establish credibility in this space. Orange should lead with its proven deployments and Gartner Magic Quadrant 2026 Leader position, while avoiding assumptions about Telefonica Tech's capabilities.

Anchored on: Private mobile networks (L0) · Colgate-Palmolive (SUP) · Mondelez International (SUP) · Gartner Magic Quadrant 2026 — 4G and 5G Private Mobile Network Services (Leader) (SUP)

What they do better: Telefonica Tech's actual capabilities in this space are unknown due to the unreadable profile.

Telia Company Telecom operator peer · structural presence

Their published position is unread — 40 page(s) fetched, none carried readable text — the site renders its content client-side. Nothing below is inferred on their behalf.

Doing here: Telia Company's published position is unread because their site renders content client-side, and none of the fetched pages carried readable text. No activity can be confirmed from their pages.

How Orange differentiates: Since Telia Company's published position is unread, Orange cannot differentiate against specific claims. However, Orange can rely on its Private mobile networks offer and manufacturing references (e.g., Saint-Gobain Glass, Skytale) to establish credibility in this space. Orange should lead with its proven deployments and Gartner Magic Quadrant 2026 Leader position, while avoiding assumptions about Telia's capabilities.

Anchored on: Private mobile networks (L0) · Saint-Gobain Glass (SUP) · Skytale (SUP) · Gartner Magic Quadrant 2026 — 4G and 5G Private Mobile Network Services (Leader) (SUP)

What they do better: Telia Company's actual capabilities in this space are unknown due to the unreadable profile.

Verizon Business Telecom operator peer · structural presence

Verizon Business presents itself as a provider of connectivity, devices, and digital solutions for small businesses, enterprises, and the public sector, emphasizing network reliability and innovation. It offers a range of services from internet and wireless plans to IoT and security solutions, targeting both commercial and government customers.

Doing here: Verizon Business's pages focus on business internet, IoT, and security solutions, but do not mention private 5G for autonomous mobile robots in manufacturing. Their IoT solutions include Sensor Insights for managing IoT ecosystems, and they offer public safety network solutions, but nothing on deterministic communications for industrial automation.

How Orange differentiates: Orange differentiates with its Private mobile networks offer, which is tailored for industrial use cases like autonomous mobile robots, and has manufacturing references such as Everise and Carso. Verizon Business's pages do not show a private 5G offering for manufacturing, focusing instead on general IoT and security. Orange can lead with its European manufacturing deployments and Gartner Magic Quadrant 2026 Leader position, while avoiding a direct comparison on US-centric solutions where Verizon has strength.

Anchored on: Private mobile networks (L0) · Everise (SUP) · Carso (SUP) · Gartner Magic Quadrant 2026 — 4G and 5G Private Mobile Network Services (Leader) (SUP)

What they do better: Verizon Business has a strong IoT and security portfolio in the US market, which Orange does not directly match.

Qualifying questions for the first meeting

- 1 Do you currently operate mobile robots or AGVs that require real-time control, and what wireless technology are they using today?
- 2 Have you experienced wireless reliability issues (e.g., dropouts, latency spikes) that have caused production stoppages or safety incidents?
- 3 Are you planning to deploy new autonomous or remote-controlled equipment in the next 12-18 months that would require deterministic low-latency connectivity?
- 4 What are your current QoS requirements for robot control (e.g., latency, jitter, reliability), and how are they measured?
- 5 Do you have an existing IT/OT integration team that could support the deployment of a private network, or would you need a managed service?
- 6 What is your current stance on network security and data sovereignty for industrial operations?

Objections you will meet

They will say	You can answer
Wi-Fi is cheaper and works fine for our current robots.	It's true that Wi-Fi is often less expensive upfront, but as you scale to more robots and more critical operations, Wi-Fi's reliability and latency issues become a bottleneck. Private 5G offers deterministic performance that Wi-Fi cannot guarantee, which can reduce downtime and improve safety.
We don't have the in-house expertise to run a private 5G network.	That's a valid concern, but Orange offers managed private mobile network services, so you don't need to build that expertise. We handle the design, deployment, and ongoing operation, allowing your team to focus on manufacturing.
We already have a private 5G pilot with another vendor.	Pilots are a great way to validate, but scaling to production requires proven reliability and support. Orange is a Leader in the Gartner Magic Quadrant for private mobile networks, and we have the certifications and expert teams to take you from pilot to full-scale deployment.

Risks and unknowns

The main risk is that the customer may not have a clear business case for remote robot control, or may be satisfied with Wi-Fi for current operations. The technology is still emerging; while demonstrations exist, large-scale industrial deployments are limited. There is uncertainty about the total cost of ownership and the operational complexity of managing a private 5G network. Additionally, the specific performance in the customer's environment (e.g., factory layout, interference) is unknown until a site survey is done.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a manufacturing prospect, highlighting Orange's Leader position in the Gartner Magic Quadrant for private mobile network services, and referencing the successful deployments at Colgate-Palmolive, Mondelez International, and Saint-Gobain Glass to demonstrate credibility.
Sales	In a meeting with a manufacturing operations executive, say: 'We've seen leading manufacturers like Colgate-Palmolive and Mondelez International deploy private 5G to modernize operations, and we can help you achieve similar outcomes with our private mobile networks.'
Strategist / Innovator	Commission a deep-dive brief on private 5G for autonomous mobile robots, leveraging Orange's Leader position in the Gartner Magic Quadrant and the Colgate-Palmolive, Mondelez International, and Saint-Gobain Glass references to validate manufacturing demand.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-B11748E56A0F	2026-06-29	2026 IEEE 12th Internatio...	1	Teleoperated Driving via a Private 5G Network: Demonstration with Remotely-Driven Robots
SIG-18012D9BCACF	2026-06-19	Springer Science and Busi...	1	Cross-Layer Latency Optimization for URLLC Services in 5G MEC Networks Using Adaptive Network S...
SIG-DDBBDA6A8C40	2026-06-19	International Scientific ...	1	Private 5G and Network Slicing for Differentiated Services: Exploring How Network Slicing Withi...
SIG-A88B3ABE1CFD	2026-06-01	cordis.europa.eu	1	Serverless Infrastructure for neXt-Generation 6G networks
SIG-A33CFB8FE031	2026-05-24	2026 IEEE International C...	1	WS14: The Fifth Workshop on Industrial Private Beyond-5G Wireless Networks, PB5GBWN
SIG-67F7CBD007F7	2026-05-22	Applied Sciences	1	Deployment and Coverage Optimization Methods for Base Stations Under Multi-Type Terminal Scenar...
SIG-24AFE29BA225	2026-05-18	openalex.org	1	Private LTE/5G, IoT, edge
SIG-0D7B53606AC6	2026-04-21	2026 IEEE 22nd Internatio...	1	A Multi-Technology Experimentation Platform for Heterogeneous Industrial IoT Communications: Fr...
SIG-FFC21218DC93	2026-03-12	openalex.org	1	Impact of 5G Latency and Jitter on TAS Scheduling in a 5G-TSN Network: An Empirical Study
SIG-4231EEBFD758	2026-01-27	openalex.org	1	Software-based 5G industrial wireless private networks :
SIG-048860EE76A9	2026-01-01	IEEE Access	1	Evolution of Industry 5.0 using AI based AMRs in Industrial Private 5G Network
SIG-058611E4CAAB	2026-01-01	openalex.org	1	TrustedSIMNet: A Reproducible Experimental Methodology for Programmable-UICC Communication over...
SIG-2373B3F43C61	2026-01-01	IEEE Wireless Communicati...	1	Quantisation-Aware RIS Placement Optimisation for AGV Connectivity in Industrial Private 5G Net...
SIG-9798339F3F2D	2026-01-01	IEEE Access	1	Multi-Connectivity for Industrial Automation: Achieving Deterministic Reliability Through Priva...
SIG-FD57D57901DD	2026-01-01	Elsevier BV	1	Multi-Objective Optimization of Latency, Reliability, and Throughput in 5G Network Slicing for ...
SIG-BC7FE33C6859	2025-11-28	openalex.org	1	Research on the Implementation of New Communication Technologies to Improve Quality and Stabili...
SIG-3056D9FEB764	2025-11-17	openalex.org	1	Data-Driven, Real-Time Diagnostics of 5G and Wi-Fi Networks Using Mobile Robotics
SIG-873E0DC9CA64	2025-11-07	International Journal of ...	1	Architecting and Analyzing Private 5G/LTE Networks for Ultra-Reliable and Deterministic Industr...
SIG-0009CF24E758	2025-10-30	openalex.org	1	AI-Native Cloud-RAN Orchestration for Enterprise Private 5G Using Digital Twin Models
SIG-77D15EBBA923	2025-10-27	openalex.org	1	QoS-Aware Proportional Fairness Scheduling for Multi-Flow 5G UEs: A Smart Factory Perspective
SIG-5E2C87F5E70E	2025-10-14	openalex.org	1	5G and UWB Integration for Robot Collaboration
SIG-D031822F4DA6	2025-09-09	openalex.org	1	Closed-Loop Network Control for Industrial Edge Computing: A Telemetry-Driven and AI-Based Appr...
SIG-D1BCC462A8B9	2025-09-09	openalex.org	1	Testbed and Software Architecture for Enhancing Security in Industrial Private 5G Networks
SIG-D9E7FFE6A4A5	2025-08-21	openalex.org	1	Head Smart - Revolutionizing Industrial Safety with IoT
SIG-10996786782A	2025-08-18	openalex.org	1	A survey on 5G private and B5G network threats and safeguarding AI-based security mechanisms th...
SIG-6FBA261C5571	2025-08-08	International Journal on ...	1	Security Challenges and Solutions in Private Wireless (LTE/5G) Network Implementations
SIG-CC7D9567F38D	2025-07-31	International Journal for...	1	Private 5G Network: Security, Challenges, and Comparison with Wi-Fi
SIG-B22C8D9EE9B5	2025-03-05	Institute of Electrical a...	1	Private 5G Cellular Networks: Transforming Tacoma Port Operations

SIG-02E085CB6CF2	2026-07-03	Telecom Review Africa	2	Private 5G and Industrial IoT: From Connectivity to Competitive Advantage - Telecom Review Afri...
SIG-08997F00E240	2026-06-16	The Fast Mode	2	Ericsson Expands Private 5G Solution with Verizon Business for Global Enterprise Deployments - ...
SIG-B58CB03F1372	2026-06-15	IoT News	2	Industrial automation drives private 5G past 2,000 deployments - IoT News
SIG-AC669DE39309	2026-05-15	Airtel	2	Private 5G Networks: Top Industries Using 5G for Business - Airtel
SIG-6CD9C9F5163D	2026-04-10	cio.com	2	Cargill deploys private 5G to aid factory AI and automation efforts - cio.com
SIG-5960AC743F0D	2026-04-09	Developing Telecoms	2	Inwi and China Mobile deploy private 5G for factory in Morocco - Developing Telecoms
SIG-1FF16F75FD01	2026-03-03	Food and Drink Digital	2	Why is Cargill Deploying Private 5G Across its Sites? - Food and Drink Digital
SIG-8B7741B3D9CF	2026-02-18	NTT Data	2	Cargill Modernizes Manufacturing Operations with NTT DATA's Private 5G Network - NTT Data
SIG-CED06E9FAEEB	2026-02-06	bing.com	2	Private LTE/5G networks reached 6,500 deployments in 2025
SIG-EE37C453692B	2025-12-01	IoT Business News	2	Private 5G for Smart Manufacturing: Lessons Learned from First Productive Deployments - IoT Bus...
SIG-F553F938CA74	2025-11-19	NTT, Inc.	2	Connected, smarter, faster: What every manufacturer wants to know about private 5G - NTT, Inc.
SIG-FC9F77C07F6F	2025-11-07	Telecom Review Asia	2	5G SA Momentum in Asia: Who's Leading the Standalone Race? - Telecom Review Asia

6 further sources are attached to this topic and are listed in the radar. Truncated here rather than silently dropped.

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-20T10:10:31+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-21T15:36:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:17+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-27T13:35:15+00:00
Pipeline	0.1.0	2026-08-25

Generated 2026-08-27 13:35 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.